Laboratory and Instrumental Approaches for Measuring Microplastic Concentration in Surface Waters

Working Principles, Evidence Map and Validation Roadmap

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Abstract

This standalone draft examines laboratory reference methods and candidate monitoring instruments for microplastic (MP) concentration in river and lake *surface water*. It draws on 10,833 preliminary include-candidate records from deduplicated Web of Science and Scopus exports, of which 6,654 have the overlapping Laboratory Methods label. Publication counts and title/abstract keyword frequencies characterise the literature database, not the number of experimentally validated devices. Selected source-matched papers illustrate the working principles and their limitations. The immediate engineering goal is to establish a reproducible chemistry-laboratory reference and then test whether portable instruments can predict reference MP concentrations under controlled and natural-water conditions.

1 Dataset and interpretation

The initial merged search contained 18,942 database records. Deduplication removed 7,318 and document-type filtering removed 727, leaving 10,897 records. Preliminary screening labelled 10,833 as include candidates and 64 for manual review. The Laboratory Methods tag covers 6,654 of the 10,833 candidates. Subcategory tags were generated by applying predefined keyword expressions to titles and abstracts within the 6,654; one record may mention multiple methods and a term mention does not confirm its use. Full-text eligibility and critical appraisal remain outstanding.

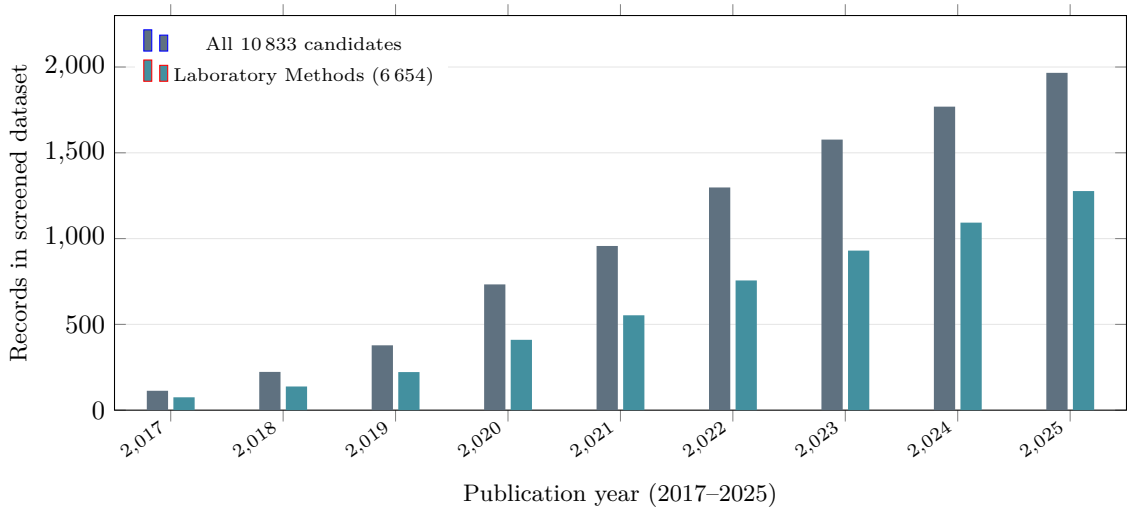


Figure 1: Annual number of records *within the screened dataset* (2017–2025). This is a descriptive publication trend, not a measure of method success or systematic-review inclusion. Data for 2026 are omitted to avoid partial-year comparison.

2 What a surface-water MP measurement actually reports

In most field surveys, a known volume of water is collected from a defined near-surface depth, retained on a filter or mesh, and analysed in the laboratory. The measurable quantity depends on *particle size, sampling depth,*

collection volume and identification criterion. Optical counting and spectral polymer identification can yield *particles per litre*, whereas thermal analysis such as pyrolysis–GC–MS can yield *polymer mass per volume*; these are different quantities and cannot be interchanged without additional size and mass information [1], [2], [3]. Small particles below the filter's effective cutoff may not be included. Surface measurements also cannot automatically represent the riverbed or the entire water column [4].

Table 1: Main working principles, output quantities and expected limitations. Examples refer to studies present in the 10,833-record candidate export.

Approach	Physical / chemical working principle	Typical output	Main validation concern
Surface sampling + filtration	Defined volume is captured; size-based retention of particles on a membrane or net [1], [5].	Volume filtered; retained particle fraction	Representative depth, flow, mesh size, recovery
Microscopy / counting	Suspected particles are segmented by morphology, illumination and size [6], [7].	Suspected particles/L	Natural debris and observer/model false positives
Micro-FTIR	Infrared absorption spectrum is matched against polymer spectra [8], [9].	Polymer-confirmed count, type, size (within method range)	Small-particle identification, spectra quality
Raman	Inelastic laser scattering identifies chemical fingerprints [8], [10].	Polymer type; confirmed particles	Fluorescence, matrix effects, time per particle
Pyrolysis–GC–MS	Thermal breakdown products identify and quantify polymers by mass [2], [3].	Polymer mass per volume	Particle number and size generally not measured
Fluorescence imaging	Fluorescent stain enhances visual contrast of particles [11].	Fluorescent suspected particle count	Non-specific staining; confirmation needed
Polarised scattering	Polarisation-resolved particle-scattered light measures optical features [12].	Particle optical signatures	Sediment/algae discrimination and range
RGB / remote imagery	Image-derived water features are linked through a model to lab reference values [13], [14].	Calibrated prediction, if validated	No direct polymer identification; site dependence

3 Research-method frequencies

Figure 2 and Table 2 show the full set of fourteen categories in the previously generated Laboratory Methods classification. They are *title/abstract keyword matches* among 6,654 candidate papers. Counts overlap and should not be used as evidence of equipment adoption, accuracy or comparative effectiveness.

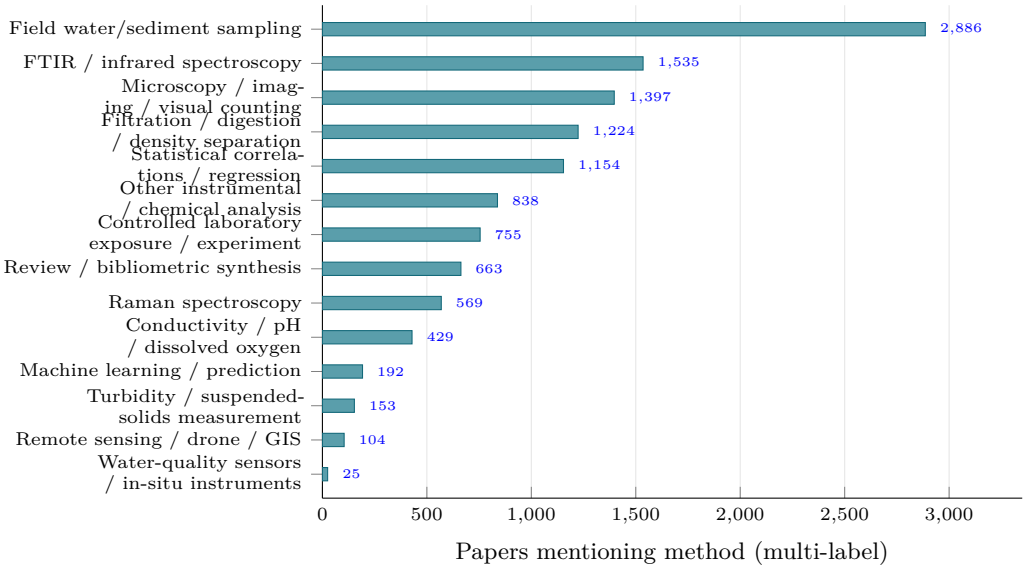


Figure 2: Automatically tagged laboratory-method keywords in the Laboratory Methods subset ($n = 6{,}654$); not confirmed study procedures.

Table 2: All defined Laboratory Methods categories in the abstract-screened subset.

Category	Mentions	Share of 6,654
Field water/sediment sampling	2,886	43.4%
FTIR / infrared spectroscopy	1,535	23.1%
Microscopy / imaging / visual counting	1,397	21.0%
Filtration / digestion / density separation	1,224	18.4%
Statistical correlations / regression	1,154	17.3%
Other instrumental / chemical analysis	838	12.6%
Controlled laboratory exposure / experiment	755	11.3%
Review / bibliometric synthesis	663	10.0%
Raman spectroscopy	569	8.6%
Conductivity / pH / dissolved oxygen	429	6.4%
Machine learning / prediction	192	2.9%
Turbidity / suspended-solids measurement	153	2.3%
Remote sensing / drone / GIS	104	1.6%
Water-quality sensors / in-situ instruments	25	0.4%

4 Representative findings relevant to equipment selection

Table 3: Selected studies in the original 10,833 candidates: procedures and results reported in their abstracts. Values are source-specific, not a pooled estimate; full-text verification remains necessary.

Study and context	Reported method / principle	Reported outcome and relevance
Odora et al. (2024), Balu River [15]	Six near-surface sampling sites; stereomicroscopy and FTIR.	Mean 102.5 ± 12.83 items/L; variations among locations reinforce the need for spatial reference samples.
Rojas-Luna et al. (2023), Luruaco Lake [5]	72 samples across dry/rainy campaigns; peroxide digestion, NaCl density separation and filtration.	Abundance 0–3.83 particles/L; sampling period and preparation affect comparison.
Chen et al. (2021), Langat River [14]	508 water samples over 12 months at eight sites; discharge and suspended-sediment associations.	Mean 4.39 particles/L; observed sediment/discharge relationships suggest *testing* turbidity as a proxy, not assuming it works everywhere.
Chen et al. (2023), Semenyih River [4]	Replicate river-water and riverbed samples at eight sites over three dates.	Water mean 3.12 ± 2.49 particles/L; sediment results differed spatially, so surface-only monitoring has defined limits.
Amsterdam canals study (2024) [2]	Surface-water pump in the 10–300 μm fraction; micro-FTIR imaging versus pyrolysis–GC–MS.	16–107 particles/m^3 in sampled size range; showed why count and mass measurement are complementary.
Raman spiking study (2024) [10]	Known plastic nanoparticles spiked into artificial and environmental fresh/saltwater; Raman analysis.	Detection limits depended on particle properties and water matrix; informs matrix-interference testing.
Liu et al. (2021) [12]	Polarisation of light scattered at 120°; machine-learning classification.	Demonstrated discrimination using test particles, algae and natural-water samples; feasibility is not field-calibrated concentration accuracy.
Lee et al. (2022) [7]	U-Net neural-network segmentation and post-processing of microscope images.	Reported human-level counting in cluttered images; still requires labelled data and chemical confirmation.

5 Selecting the laboratory reference and field device

Reference measurement. A chemistry laboratory can establish a reproducible particle-concentration reference by sampling a known water volume, retaining a specified size fraction, using contamination-controlled preparation, counting candidate particles and spectrally confirming polymer identity on a defensible sampling or subsampling basis [1], [8], [16]. Where polymer mass rather than number is central, pyrolysis–GC–MS provides a complementary chemical-mass reference rather than an interchangeable count [2], [3].

Portable device. Turbidity or RGB signals are mainly sensitive to *all* optical scatterers, including mineral sediment and organic matter. Conductivity reflects dissolved ions and therefore does not specifically identify plastic particles. A correlation with microplastic concentration must be established from *paired* laboratory measurements, with independent tests under varied matrix conditions. The Langat River study reported an observational association of MP counts with suspended sediments, while controlled flume evidence indicates that suspended sediments can also change where fibres reside in the water column [14], [17]. No universal turbidity–MP or conductivity–MP conversion is established by the screened counts.

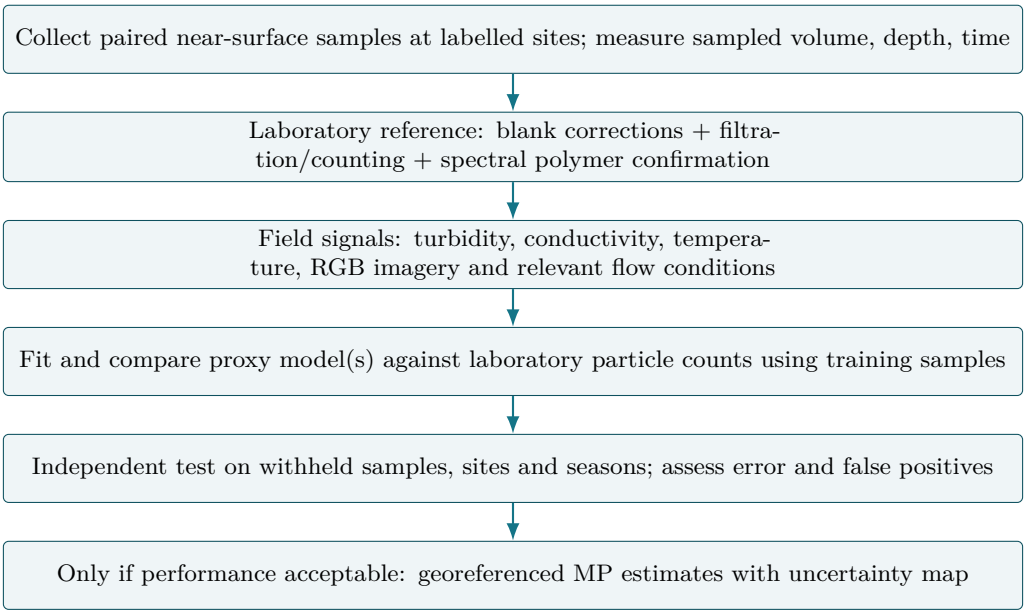


Figure 3: Proposed test-and-validation chain for the Year 3 project. This is an engineering plan, not a result established by the screened literature.

Table 4: Proposed controlled-sample matrix to discuss with the chemistry laboratory. The entries are design recommendations, not previously validated operating concentrations.

Test material	Purpose	Measurement / success criterion
Filtered-water blank and procedural blank	Identify lab-air, container and reagent contamination	Blank-corrected particle counts; document contamination events
Known MP particle spikes at multiple levels	Establish response over measurable range	Measured versus independently verified particles/L and recovery
Different particle sizes, shapes and polymers	Test dependence on material properties	Per-group false negatives and counting error
Non-plastic sediment/algae without MPs	Challenge optical specificity	False-positive signal against clean reference
Mixed MPs + sediment at variable turbidity	Assess matrix interference	Error relative to matched reference MP count
Salinity/ionic-strength and pH perturbations	Challenge conductivity / resistivity proxy	Verify whether apparent signal tracks ions instead of MPs
Independent natural river-water campaigns	Out-of-sample generalisation	MAE/RMSE, bias, site-to-site stability, confidence limits

For a particle-count reference, use a fixed sampling volume V (L), polymer-confirmed count N_{MP}, adjusted blank contribution N_{blank}, and a separately estimated recovery fraction r when justified:

$$C_{\mathrm{ref}}\,(\mathrm{particles/L}) = \frac{N_{\mathrm{MP}} - N_{\mathrm{blank}}}{V\,r}. \tag{1}$$

Record the lower size limit, sample depth, blank adjustment and any recovery correction explicitly. If r has not been measured, report uncorrected concentrations and recovery uncertainty rather than assuming $r = 1$.

Decision gate. Begin with *laboratory reference feasibility* (instrument access, size limit, available polymer standards, blank/recovery controls), then a *small controlled pilot*, then *paired river samples*. If a low-cost turbidity/conductivity model cannot predict polymer-confirmed MP counts robustly, change to a direct particle-detection or sample-concentration architecture rather than reporting proxy values as MP concentration.

6 Limitations and immediate next steps

This is a *preliminary abstract-based evidence map*. The 6,654 method-tagged records have not been confirmed as using the method, and the selected papers are illustrative rather than an exhaustive systematic synthesis. Before final thesis claims, inspect full texts for sampling depth, size cutoff, analytical blanks, recovery, polymer confirmation and uncertainty. Arrange a chemistry-lab meeting to select the reference endpoint (particles/L, optionally mass/L), source characterised reference MP particles, define a feasible sampling/filtration volume, and conduct blinded replicate tests.

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